

Feynman Prescription, Residues, and Regularization of Real Poles

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Abstract

These notes develop the Feynman prescription starting from a model integral with poles on the real axis. We first introduce the residue calculus through the Laurent series, including simple poles and poles of higher order, and study how infinitesimal detours produce contributions proportional to $\pm i\pi$ Res. We then compute the principal value using deformed contours. Finally, we introduce the $+i\epsilon$ prescription, understood as a rule that moves the poles away from the real axis, and obtain the regularized result in the usual textbook notation. The final section connects this construction with the Feynman propagator in quantum field theory.

1 Integral with poles on the real axis

Consider the integral

$$\boxed{\int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2}, \quad a \in \mathbb{R}, \quad b > 0.} \quad (1)$$

The denominator factorizes as

$$x^2 - b^2 = (x - b)(x + b), \quad (2)$$

so the function has simple poles at

$$x = b, \quad x = -b. \quad (3)$$

The problem is that both poles lie on the real axis¹, precisely on the path of integration. Therefore, before evaluating the integral we must say how the poles are detoured. That choice is what later becomes a prescription.

2 Passing to the complex plane

We extend the real variable x to a complex variable z . The integral is then associated with

$$\int dz \frac{e^{-iaz}}{z^2 - b^2}. \quad (4)$$

¹When a pole lies on the path of integration, the integral is not defined by the algebraic expression of the integrand alone; one must also specify how that pole is avoided in the complex plane.

The relevant complex function is

$$f(z) = \frac{e^{-iaz}}{z^2 - b^2} = \frac{e^{-iaz}}{(z - b)(z + b)}. \quad (5)$$

Definition 1 (Holomorphic function). We say that a complex function f is holomorphic in a region if it is complex differentiable at every point of that region. In these notes we use the term operationally: a holomorphic function is regular, with no poles or singularities, in the region under consideration.

3 General definition of the residue

If z_0 is an isolated singularity of f , then in a punctured neighborhood around z_0 the function admits a Laurent expansion²

Definition 2 (Laurent series). Let z_0 be an isolated singularity of f . A Laurent expansion of f around z_0 is a series of the form

$$f(z) = \sum_{n=-\infty}^{+\infty} c_n (z - z_0)^n. \quad (6)$$

valid in a punctured neighborhood $0 < |z - z_0| < R$.

The punctured neighborhood means that the expansion is valid in a punctured region: the singular point is excluded, but a neighborhood around it is kept. Unlike a Taylor series, a Laurent series allows negative powers of $z - z_0$. This does not contradict the fact that, if z_0 is singular, f has no Taylor series at z_0 . The Laurent series does not describe f at the singular point, but rather in the punctured neighborhood surrounding it.

In that punctured neighborhood, the function naturally splits into a regular part, formed by the terms $c_0 + c_1(z - z_0) + \dots$, and a principal part, formed by the terms with negative powers. The regular part is what would behave like a Taylor series if it could be extended to the point; the principal part is precisely what prevents f from being analytic at z_0 .

Definition 3 (Simple pole and pole of order m). We say that z_0 is a pole of order m of f if the principal part of the Laurent series of f around z_0 has the form

$$\frac{c_{-m}}{(z - z_0)^m} + \frac{c_{-(m-1)}}{(z - z_0)^{m-1}} + \dots + \frac{c_{-1}}{z - z_0}, \quad c_{-m} \neq 0. \quad (7)$$

In particular, a simple pole is a pole of order 1; near z_0 , the singular part contains only the term

$$\frac{c_{-1}}{z - z_0}, \quad c_{-1} \neq 0. \quad (8)$$

²Pierre Alphonse Laurent (1813–1854) was a French mathematician and military engineer. The series bearing his name appeared in a work submitted to the Académie des Sciences in the 1840s. Its conceptual importance is enormous: it made it possible to treat an isolated singularity not as an obstacle where analysis breaks down, but as a source of local information encoded by negative powers. In particular, the coefficient of $(z - z_0)^{-1}$ is precisely the residue.

Example 1 (The regular part and the principal part). Consider

$$f(z) = \frac{e^z}{z}. \quad (9)$$

We use the usual Taylor series of the exponential,

$$e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots. \quad (10)$$

Dividing by z , we obtain

$$f(z) = \frac{1}{z} \left(1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots \right) = \underbrace{\frac{1}{z}}_{\text{principal part}} + \underbrace{\left(1 + \frac{z}{2!} + \frac{z^2}{3!} + \cdots \right)}_{\text{regular part}}. \quad (11)$$

The principal part is responsible for the singularity. That is why it measures its nature: if there are no negative powers, the singularity is removable; if there are finitely many of them, we have a pole; if there are infinitely many, the singularity is essential.

Definition 4 (Residue). The residue of f at $z = z_0$ is the coefficient of the term $(z - z_0)^{-1}$ in the Laurent expansion of f around z_0 :

$$\boxed{\text{Res}(f, z_0) = c_{-1}}. \quad (12)$$

Example 2 (Why the coefficient c_{-1} survives). The reason why the residue is the coefficient c_{-1} appears when we integrate the Laurent series term by term around a small circle centered at z_0 . If

$$f(z) = \sum_{n=-\infty}^{+\infty} c_n (z - z_0)^n, \quad (13)$$

then formally

$$\oint_{C_\rho} f(z) dz = \sum_{n=-\infty}^{+\infty} c_n \oint_{C_\rho} (z - z_0)^n dz. \quad (14)$$

Therefore, it is enough to understand what happens to each power $(z - z_0)^n$. We take a circle C_ρ of radius ρ around z_0 , parametrized by

$$z = z_0 + \rho e^{i\theta}, \quad dz = i\rho e^{i\theta} d\theta, \quad 0 \leq \theta \leq 2\pi. \quad (15)$$

Then, for any integer n ,

$$\oint_{C_\rho} (z - z_0)^n dz = i\rho^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta. \quad (16)$$

If $n \neq -1$, the exponential completes a full turn and its net contribution cancels:

$$i\rho^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta = i\rho^{n+1} \left[\frac{e^{i(n+1)\theta}}{i(n+1)} \right]_0^{2\pi} = 0. \quad (17)$$

On the other hand, if $n = -1$, the angular dependence disappears and we get

$$\boxed{\oint_{C_\rho} \frac{dz}{z - z_0} = \int_0^{2\pi} i d\theta = 2\pi i}. \quad (18)$$

Thus, returning to the term-by-term integral of the Laurent series, all terms vanish except the term $c_{-1}(z - z_0)^{-1}$. The consequence is

$$\boxed{\oint_{C_\rho} f(z) dz = 2\pi i c_{-1} = 2\pi i \operatorname{Res}(f, z_0).} \quad (19)$$

That is why c_{-1} is exactly the local datum of the singularity that survives in contour integrals.

If z_0 is a pole of order m , then the residue can be computed with the general formula

$$\operatorname{Res}(f, z_0) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)]. \quad (20)$$

Example 3 (Why the derivative appears for a pole of higher order). The delicate point in this formula is that multiplying by $(z - z_0)^m$ removes the whole pole, but it does not directly give the residue. What one obtains first is the analytic part that remains after removing the singularity of order m .

Consider

$$f(z) = \frac{e^z}{z^2}. \quad (21)$$

Using the Taylor series of e^z , the Laurent expansion around $z = 0$ is

$$\frac{e^z}{z^2} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2!} + \frac{z}{3!} + \cdots. \quad (22)$$

The residue is not the coefficient of $1/z^2$, but the coefficient of $1/z$. Multiplying by z^2 , one obtains

$$z^2 f(z) = e^z = 1 + z + \frac{z^2}{2!} + \cdots. \quad (23)$$

The limit at $z = 0$ of this expression extracts the constant coefficient, which corresponded to the $1/z^2$ term before multiplying by z^2 . To recover the next coefficient, namely the coefficient of $1/z$, one differentiates once:

$$\operatorname{Res}\left(\frac{e^z}{z^2}, 0\right) = \lim_{z \rightarrow 0} \frac{d}{dz} \left[z^2 \frac{e^z}{z^2} \right] = \lim_{z \rightarrow 0} e^z = 1. \quad (24)$$

In particular, if the pole is simple, $m = 1$, no derivative appears and one recovers

$$\boxed{\operatorname{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z).} \quad (25)$$

This last formula is the one used in our example, because the poles at $z = \pm b$ are simple poles.³

4 Application to the example

In our case

$$f(z) = \frac{e^{-iaz}}{(z - b)(z + b)}. \quad (26)$$

³The derivative formula is useful for poles of higher order. In these notes we write it to fix the general definition, although the concrete calculations only require the case $m = 1$.

For the pole at $z = b$,

$$\text{Res}(f, b) = \lim_{z \rightarrow b} \frac{e^{-iaz}}{z + b} = \frac{e^{-iab}}{2b}. \quad (27)$$

For the pole at $z = -b$,

$$\text{Res}(f, -b) = \lim_{z \rightarrow -b} \frac{e^{-iaz}}{z - b} = \frac{e^{iab}}{-2b}. \quad (28)$$

5 Detouring a pole on the real axis

If a pole lies on the real axis, the integral is regularized by deforming the path through a small semicircle of radius ϵ around the pole. At the end one takes the limit $\epsilon \rightarrow 0^+$. The following proposition records the local rule that we will use for these detours.

Proposition 1 (Contribution of an infinitesimal detour). *Let z_0 be a simple pole of f , and let γ_ϵ be a semicircle of radius ϵ centered at z_0 . Then*

$$\lim_{\epsilon \rightarrow 0^+} \int_{\gamma_\epsilon} f(z) dz = \begin{cases} +i\pi \text{Res}(f, z_0), & \text{if the detour is counterclockwise,} \\ -i\pi \text{Res}(f, z_0), & \text{if the detour is clockwise.} \end{cases} \quad (29)$$

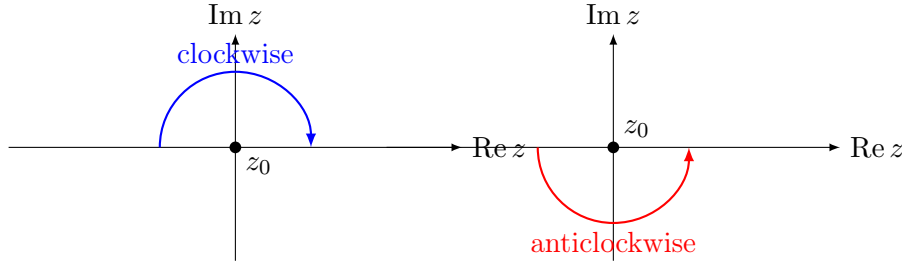


Figure 1: Infinitesimal detours around a pole on the real axis. The sign of the contribution depends on the orientation of the semicircle.

Proof. The proof is local and depends only on the singular part of f . Since z_0 is a simple pole, in a punctured neighborhood of z_0 we can write

$$f(z) = \frac{\text{Res}(f, z_0)}{z - z_0} + g(z), \quad (30)$$

where g is analytic in a full neighborhood of z_0 . In particular, g is bounded in a sufficiently small disk around z_0 : there exists a constant $M > 0$ such that

$$|g(z)| \leq M \quad (31)$$

on the semicircles γ_ϵ for ϵ sufficiently small.

We will use the elementary bound for contour integrals:⁴ if a function h satisfies $|h(z)| \leq M$ on a curve γ of length L , then

$$\left| \int_{\gamma} h(z) dz \right| \leq ML. \quad (32)$$

In our case $h = g$ and the length of a semicircle of radius ϵ is $L = \pi\epsilon$. Therefore,

$$\left| \int_{\gamma_{\epsilon}} g(z) dz \right| \leq M \text{long}(\gamma_{\epsilon}) = M\pi\epsilon. \quad (33)$$

Therefore,

$$\lim_{\epsilon \rightarrow 0^+} \int_{\gamma_{\epsilon}} g(z) dz = 0. \quad (34)$$

Thus, in the limit only the singular part $\text{Res}(f, z_0)/(z - z_0)$ contributes.

Para un rodeo anticlockwise parametrizamos

$$z = z_0 + \epsilon e^{i\theta}, \quad 0 \leq \theta \leq \pi, \quad dz = i\epsilon e^{i\theta} d\theta. \quad (35)$$

Then

$$\int_{\gamma_{\epsilon}} \frac{\text{Res}(f, z_0)}{z - z_0} dz = \int_0^{\pi} \frac{\text{Res}(f, z_0)}{\epsilon e^{i\theta}} i\epsilon e^{i\theta} d\theta = i\pi \text{Res}(f, z_0). \quad (36)$$

If the same semicircle is traversed clockwise, the parameter limits are reversed. Therefore,

$$\int_{\gamma_{\epsilon}} f(z) dz \longrightarrow -i\pi \text{Res}(f, z_0). \quad (37)$$

This proves both orientations. □

6 Residues and closed contours

Now consider a closed contour C such that f has no singularities inside it or on it. For this we have:

⁴This bound follows directly by parametrizing the curve. If $\gamma : [\alpha, \beta] \rightarrow \mathbb{C}$ is a piecewise smooth curve, then

$$\int_{\gamma} h(z) dz = \int_{\alpha}^{\beta} h(\gamma(t)) \gamma'(t) dt. \quad (I)$$

By the triangle inequality for integrals,

$$\left| \int_{\gamma} h(z) dz \right| \leq \int_{\alpha}^{\beta} |h(\gamma(t))| |\gamma'(t)| dt. \quad (II)$$

If in addition $|h(z)| \leq M$ on the whole curve, then

$$\left| \int_{\gamma} h(z) dz \right| \leq M \int_{\alpha}^{\beta} |\gamma'(t)| dt = M \text{long}(\gamma). \quad (III)$$

That is, the contour integral is bounded by “maximum of the function on the curve” times “length of the curve”. In our case, the length of the semicircle is $\pi\epsilon$, and that is why the contribution of the regular part disappears when $\epsilon \rightarrow 0^+$.

Proposition 2 (Contour with no internal singularities). *If f has no singularities inside a closed contour C , nor on the contour itself, then*

$$\boxed{\oint_C f(z) dz = 0.} \quad (38)$$

Proof. The reason is the same one that already appeared in the local calculation with Laurent series. A holomorphic function has no principal part around any point of the region: locally there are only nonnegative powers. For each regular power, $n \geq 0$,

$$(z - z_0)^n dz = d \left(\frac{(z - z_0)^{n+1}}{n+1} \right). \quad (39)$$

By the fundamental theorem of complex calculus^[5], the integral of an exact derivative over a closed path is zero. Therefore, for a regular power,

$$\oint_C (z - z_0)^n dz = \oint_C d \left(\frac{(z - z_0)^{n+1}}{n+1} \right) = 0, \quad n \geq 0. \quad (40)$$

The only term that we have already seen can survive in a closed integral is $(z - z_0)^{-1}$, whose coefficient is the residue. If there are no singularities inside C or on the contour, no coefficient c_{-1} associated with a point encircled by C appears. Therefore

$$\oint_C f(z) dz = 0. \quad (41)$$

□

Now consider a closed contour C , also with counterclockwise orientation, enclosing simple poles

$$z_1, \dots, z_N. \quad (42)$$

Choose $\epsilon > 0$ sufficiently small so that the closed disks $\overline{D_\epsilon(z_j)}$ lie inside $\text{Int}(C)$ and are pairwise disjoint. We remove those disks and call the resulting punctured region the set

$$D_\epsilon = \text{Int}(C) \setminus \bigcup_{j=1}^N D_\epsilon(z_j), \quad (43)$$

which no longer contains poles. Therefore, f is holomorphic in D_ϵ .

⁵If g has a primitive G , that is, $G'(z) = g(z)$, and $\gamma : [a, b] \rightarrow \mathbb{C}$ is a piecewise smooth curve, then

$$\int_\gamma g(z) dz = \int_a^b g(\gamma(t)) \gamma'(t) dt = \int_a^b G'(\gamma(t)) \gamma'(t) dt. \quad (\text{IV})$$

By the chain rule,

$$G'(\gamma(t)) \gamma'(t) = \frac{d}{dt} G(\gamma(t)). \quad (\text{V})$$

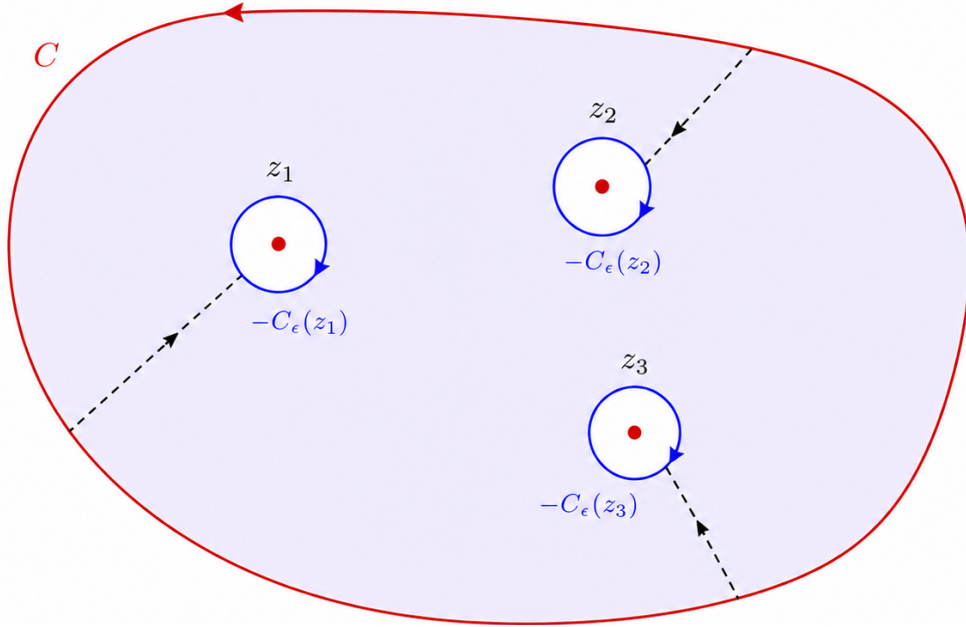
Thus,

$$\int_\gamma g(z) dz = G(\gamma(b)) - G(\gamma(a)). \quad (\text{VI})$$

If γ is closed, then $\gamma(a) = \gamma(b)$, and the integral is zero.

We now fix the orientation of the small circles in the usual way: $C_\epsilon(z_j)$ will be the circle of radius ϵ around z_j , traversed counterclockwise. This is the positive orientation around the pole, and it is the one used in the local formula $\oint_{C_\epsilon(z_j)} f(z) dz = 2\pi i \operatorname{Res}(f, z_j)$.

Now comes the orientation point. The boundary of the punctured region is not oriented piece by piece independently: it is oriented as the boundary of the whole region. One way to see this is to imagine making cuts from the outer contour to the holes. When traversing the boundary of the cut region, we start on C counterclockwise, enter through a cut, go around the hole, and return along the same cut. The two contributions from the cut cancel because they are traversed in opposite directions.



$$D_\epsilon = \operatorname{Int}(C) \setminus \bigcup_{j=1}^N D_\epsilon(z_j)$$

Figure 2: Punctured region with dotted cuts. The outer contour C is traversed counterclockwise, as indicated by the red arrow. Each circle $-C_\epsilon(z_j)$ is traversed clockwise, as indicated by the blue arrows, because it appears as the boundary of a hole of D_ϵ . Each dotted cut is traversed first from the outer boundary toward the circle and then back in the opposite direction; therefore its contributions cancel.

What remains, after canceling the cuts, is the outer contour C , with counterclockwise orientation, and each inner boundary $-C_\epsilon(z_j)$, traversed clockwise. The reason is that we are no longer orienting the small circle as an isolated contour around the pole, but as part of the boundary of D_ϵ . Thus, although $C_\epsilon(z_j)$ denotes the positive counterclockwise circle around the pole, the same circle appears in ∂D_ϵ as $-C_\epsilon(z_j)$. This orientation does not change the residue; it only records how that circle appears as a boundary of the punctured region.

We now use the previous proposition. After removing the disks $D_\epsilon(z_j)$, the poles lie outside D_ϵ . Therefore, f has no singularities in the punctured region or on its boundary, and the integral over

the full boundary vanishes:

$$\oint_{\partial D_\epsilon} f(z) dz = 0. \quad (44)$$

With this boundary orientation, we have the disjoint decomposition

$$\partial D_\epsilon = \textcolor{red}{C} \sqcup (-\textcolor{blue}{C}_\epsilon(z_1)) \sqcup \cdots \sqcup (-\textcolor{blue}{C}_\epsilon(z_N)). \quad (45)$$

Substituting this decomposition into the boundary integral, we obtain

$$0 = \oint_{\partial D_\epsilon} f(z) dz = \oint_{\textcolor{red}{C}} f(z) dz - \sum_{j=1}^N \oint_{\textcolor{blue}{C}_\epsilon(z_j)} f(z) dz. \quad (46)$$

Solving for the outer contour integral, it becomes equal to the sum of the integrals over the positive circles around the poles:

$$\oint_{\textcolor{red}{C}} f(z) dz = \sum_{j=1}^N \lim_{\epsilon \rightarrow 0^+} \oint_{\textcolor{blue}{C}_\epsilon(z_j)} f(z) dz. \quad (47)$$

Using the local contribution of each small circle,

$$\boxed{\oint_{\textcolor{red}{C}} f(z) dz = 2\pi i \sum_{j=1}^N \text{Res}(f, z_j).} \quad (48)$$

This is the form of the residue theorem that we will use.

7 Decomposition of the contour

We now evaluate the case

$$a < 0. \quad (49)$$

This condition is important because it determines in which half-plane we should close the contour. For $a < 0$, we close in the upper half-plane.

We then consider a contour C formed by four parts:

$$C = I + \gamma_{-b} + \gamma_b + \Gamma_R. \quad (50)$$

Here I represents the segment on the real axis, γ_{-b} and γ_b are the small semicircles detouring the poles at $-b$ and b , respectively, and Γ_R is the large semicircle of radius R . Afterwards we take the limit

$$R \rightarrow \infty, \quad \epsilon \rightarrow 0^+. \quad (51)$$

In this approach, the principal value is not introduced as a separate abstract definition, but as the value obtained for the real segment I after symmetrically removing the poles from the real axis and detouring them through infinitesimal semicircles in the complex plane. Those semicircles are a computational device: they allow us to close the contour and use residues. The object that we identify with the principal part is the limit of the segment I as the radii of the detours tend to zero.

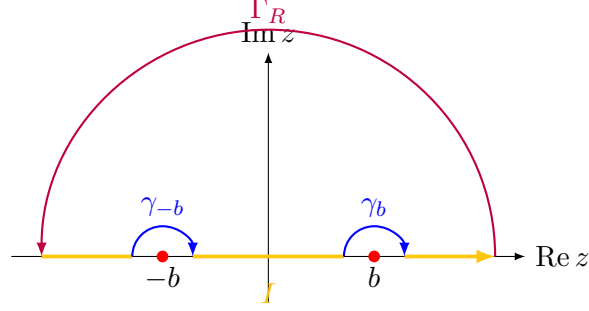


Figure 3: Upper contour with small detours above the poles $z = \pm b$. The yellow/orange segment is I , the part of the contour running along the real axis.

The reason why the closed integral vanishes is the following. Let $D_{R,\epsilon}$ be the region enclosed by the deformed contour C . The only poles of f are at $z = -b$ and $z = b$, but the small detours γ_{-b} and γ_b exclude them from $D_{R,\epsilon}$. Therefore, f has no poles inside $D_{R,\epsilon}$. By the residue theorem,

$$\oint_C f(z) dz = 2\pi i \sum_{\substack{z_j \in D_{R,\epsilon} \\ z_j \text{ pole}}} \text{Res}(f, z_j) = 2\pi i (0) = 0, \quad (52)$$

because the sum of internal residues is empty. In this case, this statement is exactly the proposition on contours without singularities applied to the deformed contour.

By the decomposition of the contour,

$$\oint_C f(z) dz = I + \int_{\gamma_{-b}} f(z) dz + \int_{\gamma_b} f(z) dz + \int_{\Gamma_R} f(z) dz. \quad (53)$$

Let us see why the large arc does not contribute. On the upper arc we write

$$z = Re^{i\theta} = R \cos \theta + iR \sin \theta, \quad 0 \leq \theta \leq \pi. \quad (54)$$

Then

$$e^{-iaz} = e^{-iaR \cos \theta} e^{aR \sin \theta}. \quad (55)$$

Since $a < 0$ and $\sin \theta \geq 0$ in the upper half-plane,

$$e^{aR \sin \theta} \longrightarrow 0 \quad (R \rightarrow \infty), \quad (56)$$

except at the endpoints of the arc. Therefore the contribution from the large arc vanishes:^[6]

$$\int_{\Gamma_R} f(z) dz \longrightarrow 0, \quad (57)$$

⁶In fact, for this integral an even simpler bound is enough. On the correct upper arc, when $a < 0$, one has

$$|e^{-iaz}| = e^{aR \sin \theta} \leq 1. \quad (\text{VII})$$

Moreover, if $R > b$, then on Γ_R

$$|z^2 - b^2| \geq ||z|^2 - b^2| = R^2 - b^2. \quad (\text{VIII})$$

Since the length of the semicircle is πR , the *ML* bound gives

$$\left| \int_{\Gamma_R} f(z) dz \right| \leq \frac{\pi R}{R^2 - b^2} \xrightarrow{R \rightarrow \infty} 0. \quad (\text{IX})$$

The exponential determines which half-plane should be chosen; the factor $1/(z^2 - b^2)$ supplies the $1/R^2$ decay that kills the length πR of the arc.

and we are left with

$$I + \int_{\gamma_{-b}} f(z) dz + \int_{\gamma_b} f(z) dz = 0. \quad (58)$$

Using the rule for the small semicircles,

$$\int_{\gamma_{\pm b}} f(z) dz = \text{sgn}(\gamma_{\pm b}) i\pi \text{Res}(f, \pm b), \quad (59)$$

where

$$\text{sgn}(\gamma_{\pm b}) = \begin{cases} +1, & \text{rodeo anticlockwise,} \\ -1, & \text{clockwise detour.} \end{cases} \quad (60)$$

Therefore, the integral over the real axis is obtained as

$$I = -i\pi \left(\text{sgn}(\gamma_{-b}) \text{Res}(f, -b) + \text{sgn}(\gamma_b) \text{Res}(f, b) \right). \quad (61)$$

Applying this result to

$$f(z) = \frac{e^{-iaz}}{z^2 - b^2}, \quad (62)$$

we use the residues

$$\text{Res}(f, b) = \frac{e^{-iab}}{2b}, \quad \text{Res}(f, -b) = -\frac{e^{iab}}{2b}. \quad (63)$$

Then

$$I = -i\pi \left[\text{sgn}(\gamma_{-b}) \left(-\frac{e^{iab}}{2b} \right) + \text{sgn}(\gamma_b) \left(\frac{e^{-iab}}{2b} \right) \right]. \quad (64)$$

In the upper contour we have chosen, both small semicircles are traversed above the pole from left to right. Therefore, both detours are clockwise:

$$\text{sgn}(\gamma_{-b}) = \text{sgn}(\gamma_b) = -1. \quad (65)$$

Then

$$I = \frac{i\pi}{2b} \left[-e^{iab} + e^{-iab} \right]. \quad (66)$$

Since

$$e^{-iab} - e^{iab} = -2i \sin(ab), \quad (67)$$

we finally obtain

$$I = \frac{\pi}{b} \sin(ab). \quad (68)$$

This is the value of the principal part of the integral.⁷ That is, for the case $a < 0$ and with the upper contour described above,

$$\boxed{\text{PV} \int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2} = \frac{\pi}{b} \sin(ab)}. \quad (69)$$

⁷The Cauchy principal value preserves the symmetry of the limit around the poles on the real axis. It is not yet the same thing as choosing a Feynman prescription; that choice appears when we move the poles away from the real axis.

At this point there is no longer any ambiguity in the signs: they have been fixed by the concrete orientation of the contour. The previous result is precisely the contribution of the main segment I , after subtracting the contributions from the infinitesimal detours around the poles.

8 Case $a > 0$: lower contour

Now we consider the complementary case

$$a > 0. \quad (70)$$

In this case it is convenient to close the contour in the lower half-plane, because the exponential factor e^{-iaz} decays on the large lower arc.

On the lower arc we write

$$z = Re^{-i\theta} = R \cos \theta - iR \sin \theta, \quad 0 \leq \theta \leq \pi. \quad (71)$$

Then

$$e^{-iaz} = e^{-iaR \cos \theta} e^{-aR \sin \theta}. \quad (72)$$

Since $a > 0$ and $\sin \theta \geq 0$,

$$e^{-aR \sin \theta} \longrightarrow 0 \quad (R \rightarrow \infty), \quad (73)$$

excepto en los extremos del arco. Therefore,^[8]

$$\int_{\Gamma_R^-} f(z) dz \longrightarrow 0. \quad (74)$$

The lower contour decomposes as

$$C_- = I + \gamma_{-b} + \gamma_b + \Gamma_R^-, \quad (75)$$

where the large arc Γ_R^- passes through the lower half-plane, while the small detours γ_{-b} and γ_b still pass above the poles. This choice keeps fixed the way in which the singularities on the real axis are avoided, even though the large arc closes below.

This contour has clockwise orientation. Moreover, with the small detours above, the poles remain inside the region enclosed by the contour. Therefore, we first evaluate the complete closed contour using the residue theorem:

$$\oint_{C_-} f(z) dz = -2\pi i [\text{Res}(f, -b) + \text{Res}(f, b)]. \quad (76)$$

⁸The same bounding argument works on the lower arc: for $a > 0$,

$$|e^{-iaz}| = e^{-aR \sin \theta} \leq 1, \quad (X)$$

and again $|z^2 - b^2| \geq R^2 - b^2$, while $\text{long}(\Gamma_R^-) = \pi R$. Therefore

$$\left| \int_{\Gamma_R^-} f(z) dz \right| \leq \frac{\pi R}{R^2 - b^2} \xrightarrow{R \rightarrow \infty} 0. \quad (XI)$$

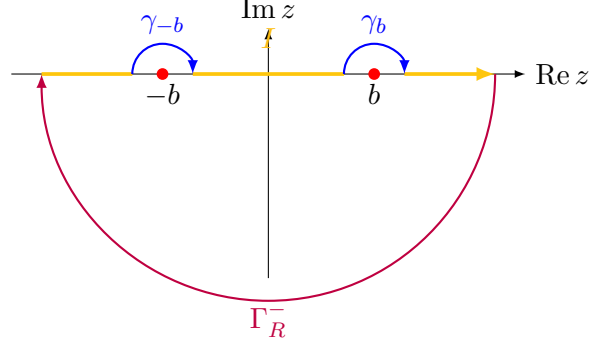


Figure 4: Lower contour for $a > 0$, with the large arc below and the small detours above the poles $z = \pm b$. The yellow/orange segment is I , the part of the contour running along the real axis.

Using

$$\text{Res}(f, b) = \frac{e^{-iab}}{2b}, \quad \text{Res}(f, -b) = -\frac{e^{iab}}{2b}, \quad (77)$$

we obtain

$$\oint_{C_-} f(z) dz = -2\pi i \left[-\frac{e^{iab}}{2b} + \frac{e^{-iab}}{2b} \right]. \quad (78)$$

Therefore,

$$\oint_{C_-} f(z) dz = -\frac{2\pi i}{2b} (e^{-iab} - e^{iab}). \quad (79)$$

When the large arc does not contribute,

$$I + \int_{\gamma_{-b}} f(z) dz + \int_{\gamma_b} f(z) dz = -2\pi i [\text{Res}(f, -b) + \text{Res}(f, b)]. \quad (80)$$

Now we compute separately the contribution of the small detours. These semicircles pass above the poles and are traversed from left to right. Therefore, both detours are clockwise:

$$\text{sgn}(\gamma_{-b}) = \text{sgn}(\gamma_b) = -1. \quad (81)$$

Thus,

$$\int_{\gamma_{-b}} f(z) dz + \int_{\gamma_b} f(z) dz = -i\pi [\text{Res}(f, -b) + \text{Res}(f, b)]. \quad (82)$$

Substituting the residues,

$$\int_{\gamma_{-b}} f(z) dz + \int_{\gamma_b} f(z) dz = -i\pi \left[-\frac{e^{iab}}{2b} + \frac{e^{-iab}}{2b} \right]. \quad (83)$$

The principal part I is obtained by subtracting from the closed contour the contribution of the two small semicircles:

$$I = \oint_{C_-} f(z) dz - \left(\int_{\gamma_{-b}} f(z) dz + \int_{\gamma_b} f(z) dz \right). \quad (84)$$

Then

$$I = -i\pi \left[-\frac{e^{iab}}{2b} + \frac{e^{-iab}}{2b} \right]. \quad (85)$$

That is,

$$I = -\frac{i\pi}{2b} (e^{-iab} - e^{iab}). \quad (86)$$

Since

$$e^{-iab} - e^{iab} = -2i \sin(ab), \quad (87)$$

one obtains

$$I = -\frac{\pi}{b} \sin(ab). \quad (88)$$

Therefore, for $a > 0$ and with the lower contour,

$$\boxed{\text{PV} \int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2} = -\frac{\pi}{b} \sin(ab)}. \quad (89)$$

Comparing with the case $a < 0$, the sign changes. Both results can be written compactly as

$$\boxed{\text{PV} \int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2} = -\frac{\pi}{b} \text{sgn}(a) \sin(ab), \quad a \neq 0.} \quad (90)$$

This solves the initial problem in the principal value sense: the model integral is interpreted as the segment I of the contour, namely the part running along the real axis. By detouring the poles and isolating the contributions from the small semicircles, we obtain a precise value for that segment I . In the next section we change viewpoint: instead of detouring poles that remain on the real axis, we introduce the Feynman prescription, which moves the poles away from the real axis and directly fixes how the integral must be interpreted.

It is important not to confuse these two procedures. The principal value is a symmetric way of interpreting the original integral. The Feynman prescription, by contrast, temporarily replaces the integrand by a regularized function with displaced poles. Thus the Feynman result is not merely the principal value: it contains a real part, which reproduces the principal value, and an imaginary part, which encodes the choice of prescription.

9 Feynman regularization

The problem with the original integral

$$\int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2} \quad (91)$$

is that the poles lie exactly on the real axis. Instead of working directly with that singular expression, the Feynman prescription introduces a new regularized function whose poles are no longer on the real axis. That is, from this point on we are not literally computing the same singular function, but a family of functions f_ϵ whose limit $\epsilon \rightarrow 0^+$ defines the prescription.

Definition 5 (Feynman prescription). The Feynman prescription consists in displacing the poles infinitesimally away from the real axis by means of a parameter $\epsilon > 0$, carrying out the calculation with the regularized function, and taking the limit $\epsilon \rightarrow 0^+$ at the end. In the case studied here, we choose

$$b \mapsto b - i\epsilon, \quad -b \mapsto -b + i\epsilon,$$

so that the right pole lies below the real axis and the left pole lies above it.

We take, for example,

$$f_\epsilon(z) = \frac{e^{-iaz}}{(z - b + i\epsilon)(z + b - i\epsilon)}, \quad \epsilon > 0. \quad (92)$$

The poles of f_ϵ are

$$z = b - i\epsilon, \quad z = -b + i\epsilon. \quad (93)$$

That is, the pole on the right is shifted slightly downward, while the pole on the left is shifted slightly upward.

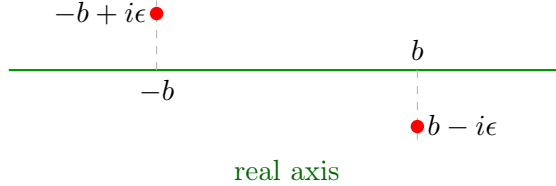


Figure 5: Displacement of the poles in the Feynman prescription.

Expanding the denominator,

$$(z - b + i\epsilon)(z + b - i\epsilon) = z^2 - b^2 + 2ib\epsilon + \epsilon^2. \quad (94)$$

Since ϵ is infinitesimal, the term ϵ^2 is neglected:

$$(z - b + i\epsilon)(z + b - i\epsilon) \simeq z^2 - b^2 + 2ib\epsilon. \quad (95)$$

Therefore, in this example we must continue working with

$$\frac{1}{z^2 - b^2 + i2b\epsilon}. \quad (96)$$

The factor $2b$ does not disappear: since b is finite, $2b\epsilon$ is still infinitesimal, but in this calculation we keep it explicitly. This is the precise form of the regularization we have constructed:

$$\frac{1}{z^2 - b^2} \longrightarrow \frac{1}{z^2 - b^2 + i2b\epsilon}. \quad (97)$$

Sometimes, in a more compact notation, one again calls the infinitesimal $2b\epsilon$ by the name ϵ , and then simply writes $+i\epsilon$. But that abbreviation is only a renaming of the infinitesimal. In this derivation we keep explicitly the regularized denominator $z^2 - b^2 + i2b\epsilon$.

9.1 Residues of the regularized function

Now we compute the residues of

$$\boxed{f_\epsilon(z) = \frac{e^{-iaz}}{z^2 - b^2 + i2b\epsilon}, \quad z_- = -b + i\epsilon, \quad z_+ = b - i\epsilon.} \quad (98)$$

whose poles are given to first order in ϵ .

For the pole $z_- = -b + i\epsilon$,

$$\text{Res}(f_\epsilon, z_-) = \lim_{z \rightarrow -b + i\epsilon} (z - (-b + i\epsilon)) \frac{e^{-iaz}}{z^2 - b^2 + i2b\epsilon}. \quad (99)$$

Evaluating the exponential factor at the pole,

$$\text{Res}(f_\epsilon, z_-) = e^{-ia(-b+i\epsilon)} \lim_{z \rightarrow -b+i\epsilon} \frac{z - (-b + i\epsilon)}{z^2 - b^2 + i2b\epsilon}. \quad (100)$$

Since

$$z^2 - b^2 + i2b\epsilon \simeq (z - (-b + i\epsilon))(z - (b - i\epsilon)), \quad (101)$$

one obtains

$$\text{Res}(f_\epsilon, z_-) = \frac{e^{-ia(-b+i\epsilon)}}{(-b + i\epsilon) - (b - i\epsilon)}. \quad (102)$$

Therefore,

$$\text{Res}(f_\epsilon, z_-) = \frac{e^{iab+a\epsilon}}{-2b + 2i\epsilon}. \quad (103)$$

Taking $\epsilon \rightarrow 0^+$,

$$\boxed{\text{Res}(f_\epsilon, -b + i\epsilon) \longrightarrow -\frac{e^{iab}}{2b}}. \quad (104)$$

Analogously, for $z_+ = b - i\epsilon$,

$$\text{Res}(f_\epsilon, z_+) = \lim_{z \rightarrow b - i\epsilon} (z - (b - i\epsilon)) \frac{e^{-iaz}}{z^2 - b^2 + i2b\epsilon}. \quad (105)$$

Using the same factorization to first order,

$$\text{Res}(f_\epsilon, z_+) = \frac{e^{-ia(b-i\epsilon)}}{(b - i\epsilon) - (-b + i\epsilon)} = \frac{e^{-iab-a\epsilon}}{2b - 2i\epsilon}. \quad (106)$$

Thus, in the limit $\epsilon \rightarrow 0^+$,

$$\boxed{\text{Res}(f_\epsilon, b - i\epsilon) \longrightarrow \frac{e^{-iab}}{2b}}. \quad (107)$$

These are the same residues that appeared for the nonregularized function, but now the poles are no longer on the real axis.

9.2 Contour integral for $a < 0$

Now consider the regularized integral

$$I_\epsilon = \int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2 + i2b\epsilon}. \quad (108)$$

For $a < 0$, the exponential factor decays when the contour is closed in the upper half-plane. Indeed, if

$$z = Re^{i\theta}, \quad 0 \leq \theta \leq \pi, \quad (109)$$

then

$$e^{-iaz} = e^{-iaR \cos \theta} e^{aR \sin \theta}. \quad (110)$$

Since $a < 0$ and $\sin \theta \geq 0$, the integral over the large arc vanishes as $R \rightarrow \infty$.

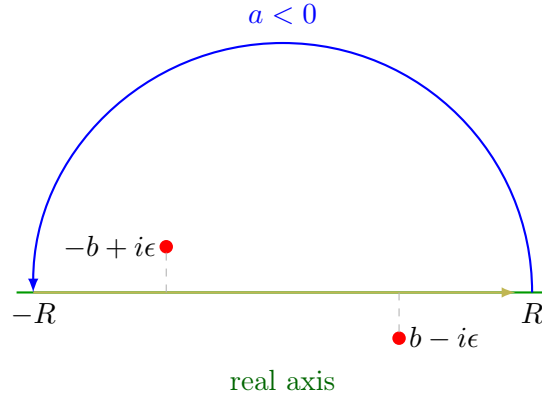


Figure 6: For $a < 0$ we close above; only the pole $-b + i\epsilon$ lies inside the contour.

The closed contour has positive orientation. By the residue theorem,

$$\oint_C dz f_\epsilon(z) = 2\pi i \operatorname{Res}(f_\epsilon, -b + i\epsilon). \quad (111)$$

Since el arco grande no contribuye,

$$I_\epsilon = 2\pi i \operatorname{Res}(f_\epsilon, -b + i\epsilon). \quad (112)$$

Using the residue computed above,

$$I_\epsilon \xrightarrow{\epsilon \rightarrow 0^+} 2\pi i \left(-\frac{e^{iab}}{2b} \right). \quad (113)$$

Therefore,

$$\boxed{I_F(a < 0) = -\frac{\pi i}{b} e^{iab}}. \quad (114)$$

Writing $e^{iab} = \cos(ab) + i \sin(ab)$, we obtain

$$I_F(a < 0) = -\frac{\pi i}{b} \cos(ab) + \frac{\pi}{b} \sin(ab). \quad (115)$$

Thus,

$$\operatorname{Re} I_F(a < 0) = \frac{\pi}{b} \sin(ab), \quad (116)$$

which agrees with the value of the principal part computed previously for $a < 0$. The Feynman prescription also adds an imaginary part:

$$\operatorname{Im} I_F(a < 0) = -\frac{\pi}{b} \cos(ab). \quad (117)$$

9.3 Contour integral for $a > 0$

For $a > 0$, the exponential factor decays when the contour is closed in the lower half-plane. If we parametrize the large arc as

$$z = Re^{-i\theta}, \quad 0 \leq \theta \leq \pi, \quad (118)$$

then

$$e^{-iaz} = e^{-iaR\cos\theta} e^{-aR\sin\theta}. \quad (119)$$

Since $a > 0$ and $\sin\theta \geq 0$, the integral over the lower arc vanishes as $R \rightarrow \infty$.

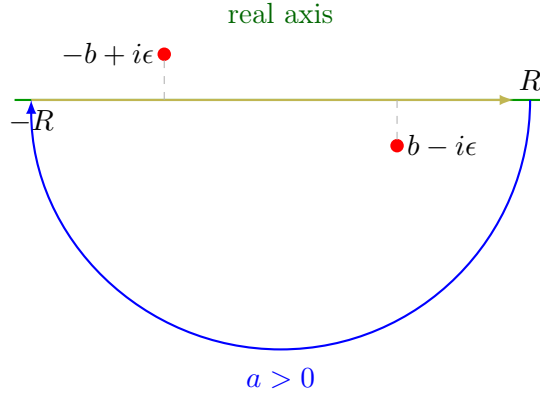


Figure 7: For $a > 0$ we close below; only the pole $b - i\epsilon$ lies inside the contour.

Now the closed contour has negative orientation, that is, clockwise. By the residue theorem,

$$\oint_{C_-} dz f_\epsilon(z) = -2\pi i \operatorname{Res}(f_\epsilon, b - i\epsilon). \quad (120)$$

Since el arco grande no contribuye,

$$I_\epsilon = -2\pi i \operatorname{Res}(f_\epsilon, b - i\epsilon). \quad (121)$$

Using the residue computed above,

$$I_\epsilon \xrightarrow{\epsilon \rightarrow 0^+} -2\pi i \left(\frac{e^{-iab}}{2b} \right). \quad (122)$$

Therefore,

$$\boxed{I_F(a > 0) = -\frac{\pi i}{b} e^{-iab}}. \quad (123)$$

Writing $e^{-iab} = \cos(ab) - i \sin(ab)$, we obtain

$$I_F(a > 0) = -\frac{\pi i}{b} \cos(ab) - \frac{\pi}{b} \sin(ab). \quad (124)$$

Thus,

$$\operatorname{Re} I_F(a > 0) = -\frac{\pi}{b} \sin(ab), \quad (125)$$

which agrees with the value of the principal part computed previously for $a > 0$. Moreover,

$$\operatorname{Im} I_F(a > 0) = -\frac{\pi}{b} \cos(ab). \quad (126)$$

The two cases are summarized as

$$\boxed{I_F(a) = -\frac{\pi i}{b} e^{-i|a|b}, \quad a \neq 0} \quad (127)$$

Its real part reproduces the principal value:

$$\boxed{\operatorname{Re} I_F(a) = -\frac{\pi}{b} \operatorname{sgn}(a) \sin(ab).} \quad (128)$$

But $I_F(a)$ does not coincide with the full principal value: it coincides only in its real part. The difference is the imaginary part,

$$\operatorname{Im} I_F(a) = -\frac{\pi}{b} \cos(ab), \quad (129)$$

which appears because the $+i\epsilon$ prescription specifies on which side of the real axis each pole lies.

The prescription does not mean that a finite physical correction is added.^[9] It means that we have defined how to avoid the poles: one lies above the real axis and the other below it.

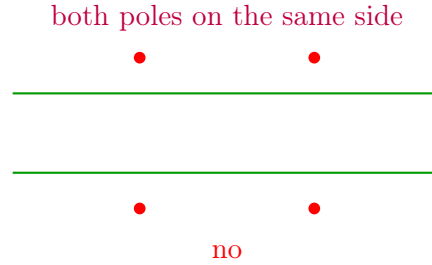


Figure 8: The Feynman prescription does not place both poles on the same side of the real axis.

9.4 Usual textbook notation

In our example the denominator appeared explicitly as

$$z^2 - b^2 + i 2b\epsilon. \quad (130)$$

⁹The limit $\epsilon \rightarrow 0^+$ is taken at the end. Before that limit, ϵ works as an instruction for defining the contour, not as a new physical parameter of the system.

Since $b > 0$ es finito, podemos definir un nuevo infinitesimal

$$\eta = 2b\epsilon, \quad \eta \rightarrow 0^+. \quad (131)$$

Then the same prescription can be written as

$$z^2 - b^2 + i 2b\epsilon = z^2 - b^2 + i\eta. \quad (132)$$

Finally, by convention, one again calls the infinitesimal η by the name ϵ . This gives the usual form

$$\boxed{\frac{1}{z^2 - b^2} \longrightarrow \frac{1}{z^2 - b^2 + i\epsilon}, \quad \epsilon \rightarrow 0^+}. \quad (133)$$

This is the compact form found in textbooks: the symbol $+i\epsilon$ does not keep the finite factors multiplying the infinitesimal, but only the essential information of the prescription.

With this notation, the final result we found is written as

$$\boxed{\int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2 + i\epsilon} = -\frac{\pi i}{b} e^{-ib|a|}, \quad a \neq 0}. \quad (134)$$

Equivalently, case by case,

$$\boxed{\int_{-\infty}^{+\infty} dx \frac{e^{-iax}}{x^2 - b^2 + i\epsilon} = \begin{cases} -\frac{\pi i}{b} e^{iab}, & a < 0, \\ -\frac{\pi i}{b} e^{-iab}, & a > 0. \end{cases}} \quad (135)$$

The compact expression appears because

$$\begin{cases} a = -|a|, & a < 0, \\ a = |a|, & a > 0, \end{cases} \quad \implies \quad e^{iab}|_{a<0} = e^{-ib|a|}, \quad e^{-iab}|_{a>0} = e^{-ib|a|}. \quad (136)$$

If the factor is absorbed before computing the poles, the result does not change. Indeed, starting from

$$z^2 - b^2 + i\eta = 0, \quad (137)$$

the poles satisfy

$$z = \pm \sqrt{b^2 - i\eta}. \quad (138)$$

For $\eta \ll 1$,

$$\sqrt{b^2 - i\eta} = b \sqrt{1 - \frac{i\eta}{b^2}} \simeq b \left(1 - \frac{i\eta}{2b^2} \right) = b - \frac{i\eta}{2b}. \quad (139)$$

Therefore,

$$z_+ = b - \frac{i\eta}{2b}, \quad z_- = -b + \frac{i\eta}{2b}. \quad (140)$$

If we now use $\eta = 2b\epsilon$, we recover exactly

$$z_+ = b - i\epsilon, \quad z_- = -b + i\epsilon. \quad (141)$$

The residues do not change in the limit either. For example,

$$\text{Res} \left(\frac{e^{-iaz}}{z^2 - b^2 + i\eta}, z_- \right) = \frac{e^{-iaz_-}}{z_- - z_+} \xrightarrow{\eta \rightarrow 0^+} -\frac{e^{iab}}{2b}, \quad (142)$$

and analogously

$$\text{Res} \left(\frac{e^{-iaz}}{z^2 - b^2 + i\eta}, z_+ \right) \xrightarrow{\eta \rightarrow 0^+} \frac{e^{-iab}}{2b}. \quad (143)$$

Thus, absorbing the factor $2b$ before or after the calculation leads to the same residues and the same regularized integral. The only thing that must not be lost is the sign of the imaginary part, which indicates on which side of the real axis each pole lies.

10 Conclusion: connection with the Feynman propagator

The calculation above shows the precise role of the $+i\epsilon$ term: it moves the poles away from the real axis and fixes which contour must be used when evaluating the integral. It does not represent a finite correction to the denominator, but rather a prescription for defining the integral in the complex plane.

This is exactly the idea that appears in quantum field theory. For a scalar field, the two-point Feynman propagator^[10] is written as

$$\Delta_F(x - y) = i \int \frac{d^4k}{(2\pi)^4} \frac{e^{-ik \cdot (x-y)}}{k^2 - m^2 + i\epsilon}. \quad (144)$$

The denominator

$$k^2 - m^2 + i\epsilon \quad (145)$$

must be read with the same logic as in the one-dimensional example: the $+i\epsilon$ indicates how the poles in the variable k^0 are shifted. That choice defines the integration contour and therefore selects the Feynman propagator.

¹⁰The two-point propagator is the Green function that encodes the amplitude for propagation between two spacetime points. The $+i\epsilon$ prescription selects the causal boundary conditions corresponding to the Feynman propagator.